
PH102: Physics-2

Electrodynamics

Lecture # 02

Line, Surface and Volume Integrals

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Previous lecture

- ✓ The notations **A**, **B**, **C** etc. are used for usual vectors which represent a meaningful physical quantity called a vector field.
- ✓ The notations *f*, *g*, *h*, *T* etc. (in *italics* and **not bold**) are used for a meaningful physical quantity called a scalar field.
- ✓ ∇ is an operator and is not a usual vector.
- ✓ The operator operates on a scalar or a vector field and generates either a scalar or a vector field depending upon a valid operation of the form ∇T , $\nabla \cdot \mathbf{v}$ or $\nabla \times \mathbf{v}$.
- ✓ The vector identities need not necessarily be always true for the operators.

Second derivatives

∇T , $\nabla \cdot \mathbf{v}$ and $\nabla \times \mathbf{v}$ are the first derivatives and are useful to generate second derivatives. The following possibilities exist with the first derivatives:

Gradient ∇T (a vector)

A) Divergence of gradient: $\nabla \cdot (\nabla T)$. (The result is a Laplacian, i.e. $\nabla^2 T$)

$$\nabla \cdot (\nabla T) = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \right) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}.$$

B) Curl of gradient: $\nabla \times (\nabla T)$.

The Curl of a gradient is zero because of the equality of cross derivatives,

$$\frac{\partial}{\partial x} \left(\frac{\partial T}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial x} \right), \quad \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial z} \right) = \frac{\partial}{\partial z} \left(\frac{\partial T}{\partial y} \right), \quad \frac{\partial}{\partial z} \left(\frac{\partial T}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial T}{\partial z} \right)$$

Caution !!! $\nabla \times (\nabla T) \neq (\nabla \times \nabla) T$ because ∇ is an operator and does not "multiply" in the usual way.

(contd...)

Divergence $\nabla \cdot \mathbf{v}$ (a scalar)

C) Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$.

Curl $\nabla \times \mathbf{v}$ (a vector)

D) Divergence of curl: $\nabla \cdot (\nabla \times \mathbf{v})$. **(the divergence of a curl is zero)**

E) Curl of curl: $\nabla \times (\nabla \times \mathbf{v})$.

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}$$

Useful to obtain the electromagnetic wave equation.

Laplacian of a vector is $(\nabla \cdot \nabla)\mathbf{v} = \nabla^2 \mathbf{v} \equiv (\nabla^2 v_x)\hat{\mathbf{x}} + (\nabla^2 v_y)\hat{\mathbf{y}} + (\nabla^2 v_z)\hat{\mathbf{z}}$.

Caution !!! $\nabla^2 \mathbf{v} = (\nabla \cdot \nabla)\mathbf{v} \neq \nabla(\nabla \cdot \mathbf{v})$.

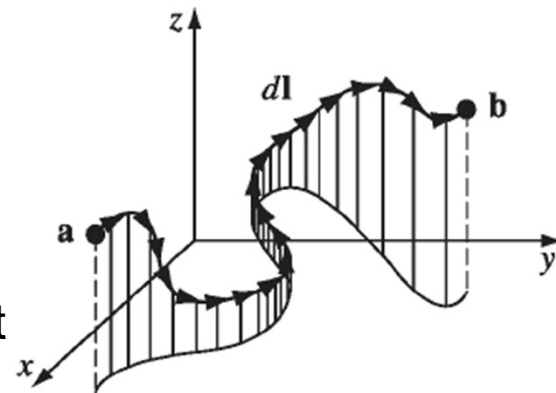
Integral Calculus: Line integral (path integral)

A line integral is an extension of an idea of integration of one variable $\int f(x)dx$ to a scalar and a vector field on any path.

It is an expression of the form $\int_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} \equiv \int_a^b \mathbf{v} \cdot d\mathbf{l}$

$\mathbf{v} \rightarrow$ a vector field,

$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$. $\rightarrow$ the infinitesimal displacement vector.



Integral is carried out along a prescribed path $\mathcal{P}$ from point $\mathbf{a}$ to point $\mathbf{b}$.

$\mathbf{v} \cdot d\mathbf{l}$ is performed at each point on the path to the next point on the path.

If the path in question forms a closed loop (that is, if $\mathbf{b} = \mathbf{a}$), put a circle on the integral sign: $\oint \mathbf{v} \cdot d\mathbf{l}$.

Usually, the value of a line integral depends on the particular path taken from $\mathbf{a}$ to $\mathbf{b}$.

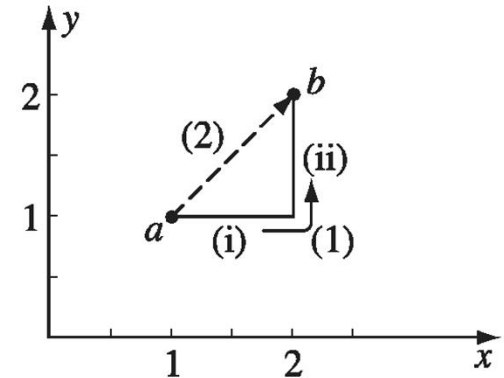
For some special vector fields, the line integral is independent of the path, and is determined entirely by the end points. A vector field having this property is conservative in nature.

Solved example 1

Calculate the line integral of the function

$$\mathbf{v} = y^2 \hat{\mathbf{x}} + 2x(y + 1) \hat{\mathbf{y}}$$

from the point $\mathbf{a} = (1, 1, 0)$ to the point $\mathbf{b} = (2, 2, 0)$, along the paths (1) and (2).



As always, $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$. Path (1) consists of two parts. Along the “horizontal” segment, $dy = dz = 0$, so

$$(i) \quad d\mathbf{l} = dx \hat{\mathbf{x}}, \quad y = 1, \quad \mathbf{v} \cdot d\mathbf{l} = y^2 dx = dx, \quad \text{so} \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_1^2 dx = 1.$$

On the “vertical” stretch, $dx = dz = 0$, so

$$(ii) \quad d\mathbf{l} = dy \hat{\mathbf{y}}, \quad x = 2, \quad \mathbf{v} \cdot d\mathbf{l} = 2x(y + 1) dy = 4(y + 1) dy, \quad \text{so}$$

$$\int \mathbf{v} \cdot d\mathbf{l} = 4 \int_1^2 (y + 1) dy = 10.$$

By path (1), then,

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{v} \cdot d\mathbf{l} = 1 + 10 = 11.$$

(contd...)

Meanwhile, on path (2) $x = y$, $dx = dy$, and $dz = 0$, so
 $d\mathbf{l} = dx \hat{\mathbf{x}} + dx \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{l} = x^2 dx + 2x(x + 1) dx = (3x^2 + 2x) dx$,
and

$$\int_a^b \mathbf{v} \cdot d\mathbf{l} = \int_1^2 (3x^2 + 2x) dx = (x^3 + x^2) \Big|_1^2 = 10.$$

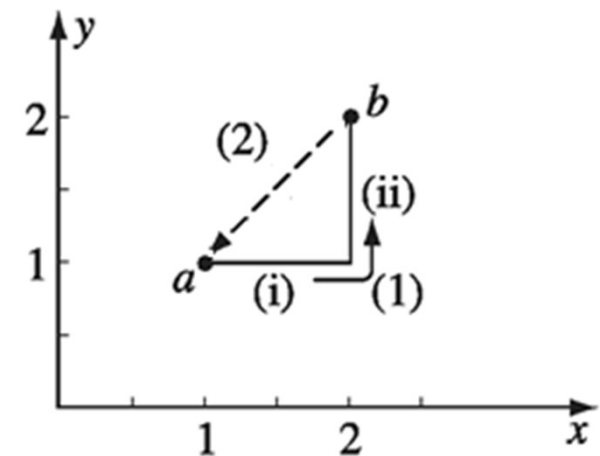
What is $\oint \mathbf{v} \cdot d\mathbf{l}$ for the loop that goes from **a** to **b** along (1) and returns to **a** along (2)?

Along path (2)

$$\int_a^b \mathbf{v} \cdot d\mathbf{l} = \int_2^1 (3x^2 + 2x) dx = -10.$$

For the loop that goes *out* (1) and *back* (2), then,

$$\oint \mathbf{v} \cdot d\mathbf{l} = 11 - 10 = 1.$$



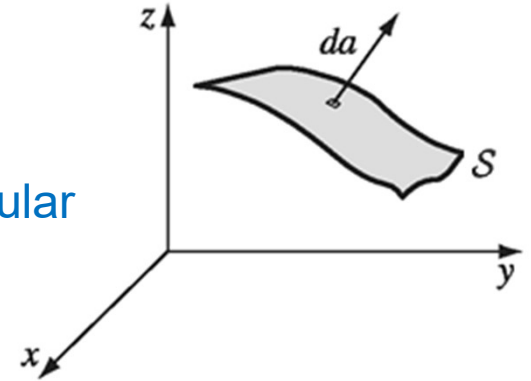
NOTE: Always use $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$. Let the limits take care of the components of the displacement either along positive or negative direction.

Integral Calculus: Surface integral

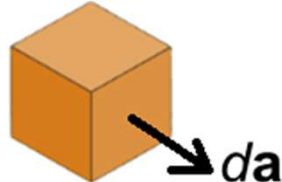
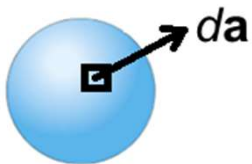
A surface integral is an expression of the form $\int_S \mathbf{v} \cdot d\mathbf{a}$.

$\mathbf{v} \rightarrow$ a vector field,

$d\mathbf{a} \rightarrow$ the infinitesimal patch of area, with direction perpendicular to the surface S .



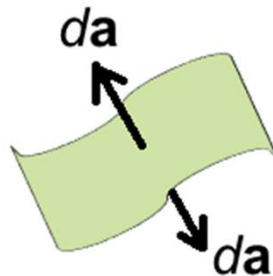
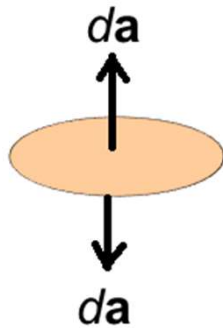
For a closed surface, a circle is marked on the integration sign, $\oint \mathbf{v} \cdot d\mathbf{a}$.



Closed surface (sphere, cube etc.)

Direction for $d\mathbf{a}$: outward normal is positive.

For open surfaces, the direction for $d\mathbf{a}$ is arbitrary (normal to top or bottom surfaces).



A plane surface is characterized by a boundary at the edges.

(contd...)

To further specify the area for an open surface, we draw arrows that run around the perimeter of the area in one of the two possible directions.

Area vector will point in the direction following the right hand thumb rule.



Only a **planar area** can be **represented as a single vector**. **Non-planar** areas like a hemisphere **can not be represented by a single vector** (it has a curved surface and a flat surface).

The value of a surface integral generally depends on the particular surface chosen, but there is a special class of vector fields for which it is independent of the surface, *and is determined entirely by the boundary*.

Elementary area on a surface

Let S be a smooth surface: $z = f(x, y)$.

Project it on xy -plane: R be the projection.

Choose an elementary area da on S and let $\hat{n}$ be a unit vector perpendicular to it.

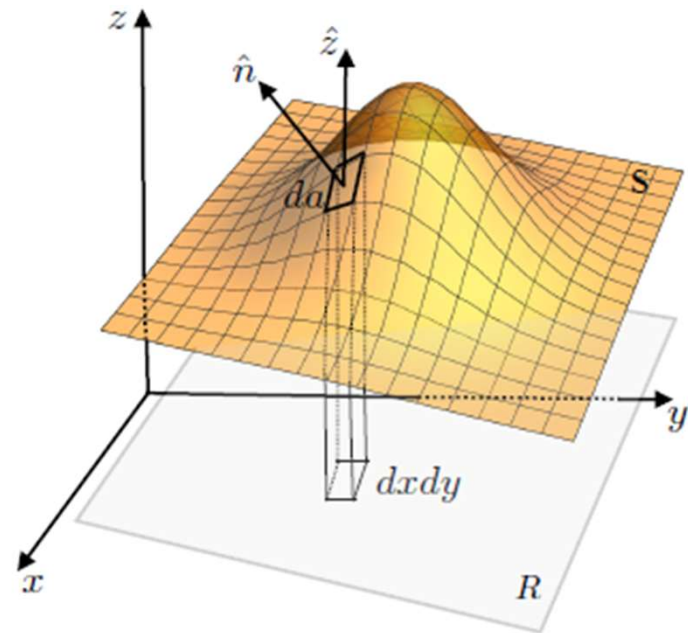
Projection of da on xy -plane is $dxdy$.

$$\therefore dxdy = |\hat{n} \cdot \hat{z}| da$$

$$da = \frac{dxdy}{|\hat{n} \cdot \hat{z}|}$$

Hence we can denote da as vector area

$$d\vec{a} = \left(\frac{dxdy}{|\hat{n} \cdot \hat{z}|} \right) \hat{n} = \hat{n} da$$

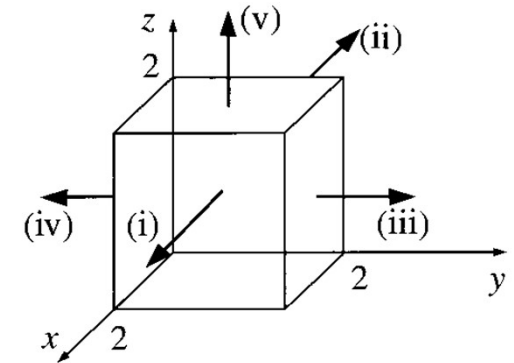


Solved example 2

Calculate the surface integral of

$$\mathbf{v} = 2xz \hat{\mathbf{x}} + (x+2) \hat{\mathbf{y}} + y(z^2-3) \hat{\mathbf{z}}$$

over five sides (excluding the bottom) of the cubical box (of side 2 unit). Let "upward and outward" be the positive direction, as indicated by the arrows.



(i) $x = 2$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = 2xz dy dz = 4z dy dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 4 \int_0^2 dy \int_0^2 z dz = 16.$$

(ii) $x = 0$, $d\mathbf{a} = -dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = -2xz dy dz = 0$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 0.$$

(iii) $y = 2$, $d\mathbf{a} = dx dz \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = (x+2) dx dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 (x+2) dx \int_0^2 dz = 12.$$

(iv) $y = 0$, $d\mathbf{a} = -dx dz \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = -(x+2) dx dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = - \int_0^2 (x+2) dx \int_0^2 dz = -12.$$

(v) $z = 2$, $d\mathbf{a} = dx dy \hat{\mathbf{z}}$, $\mathbf{v} \cdot d\mathbf{a} = y(z^2-3) dx dy = y dx dy$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 dx \int_0^2 y dy = 4.$$

The total flux is $\int_{\text{surface}} \mathbf{v} \cdot d\mathbf{a} = 16 + 0 + 12 - 12 + 4 = 20$.

Integral Calculus: Volume integral

A volume integral is an expression of the form $\int_V T d\tau$.

$T \rightarrow$ a scalar field,

$d\tau = dx dy dz \rightarrow$ an infinitesimal volume element.

If T is the density of a substance (which might vary from point to point), then the volume integral would give the total mass.

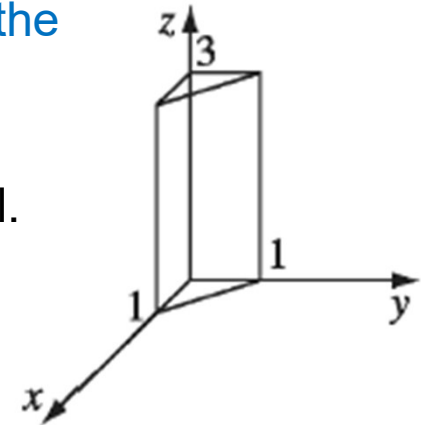
The volume integral of a vector function is given by

$$\int \mathbf{v} d\tau = \int (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) d\tau = \hat{\mathbf{x}} \int v_x d\tau + \hat{\mathbf{y}} \int v_y d\tau + \hat{\mathbf{z}} \int v_z d\tau$$

The unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are constants and therefore they are taken outside the integral.

Solved example 3

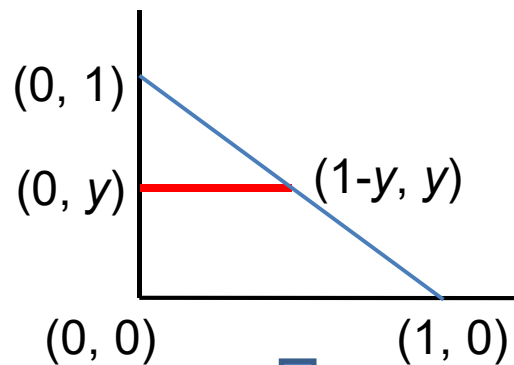
Calculate the volume integral of the function $T = xyz^2$ over the prism in Figure.



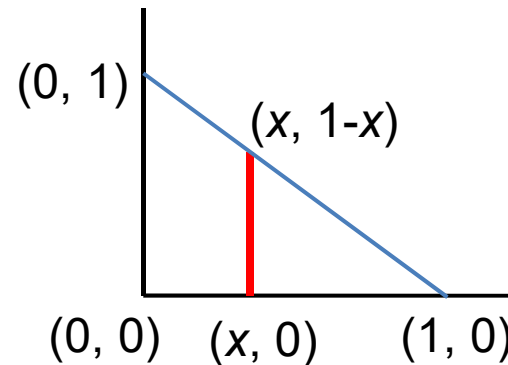
z is independent of x and y and hence is an independent integral.

x and y are dependent on each other by the relation: $x + y = 1$.

Integration on x and y can be carried out in two different ways:



OR



x runs from 0 to $(1-y)$; then y goes from 0 to 1; and z from 0 to 3.

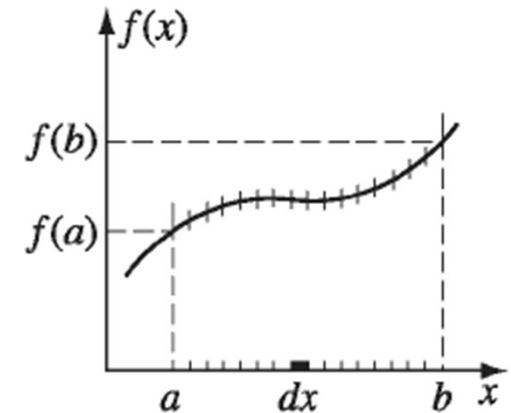
$$\int T d\tau = \int_0^3 z^2 \left\{ \int_0^1 y \left[\int_0^{1-y} x dx \right] dy \right\} dz = \frac{1}{2} \int_0^3 z^2 dz \int_0^1 (1-y)^2 y dy = \frac{1}{2} (9) \left(\frac{1}{12} \right) = \frac{3}{8}.$$

The Fundamental Theorem of Calculus

Suppose $f(x)$ is a function of one variable. **The fundamental theorem of calculus states:**

$$\int_a^b \left(\frac{df}{dx} \right) dx = f(b) - f(a).$$

or,
$$\int_a^b F(x) dx = f(b) - f(a), \text{ where, } F(x) = df/dx.$$



The *integral* of a *derivative* over some *region* is given by the *value of the function* at the end points (*boundaries*).

The fundamental theorem says that if you chop the interval from a to b into many tiny pieces, dx , and add up the increments df from each little piece, the result is equal to the total change in f : $f(b) - f(a)$.

There are two ways to determine the total change in the function:

- (a) subtract the values at the ends, or
- (b) go step-by-step, adding up all the tiny increments as you go.

You'll get the same answer either way.

The Fundamental Theorem for Gradients

If a scalar function of three variables $T(x, y, z)$ changes by a small amount dT after going through an infinitesimal displacement $d\mathbf{l}_1$, starting from a point $\mathbf{a}$, then

$$dT = (\nabla T) \cdot d\mathbf{l}_1.$$

Now, we move a little further, by an additional small displacement $d\mathbf{l}_2$, and then to $d\mathbf{l}_3$ and so on till we reach a point $\mathbf{b}$, but by calculating the incremental change in T (i.e., dT) at each of the incremental displacement. *The total change in T in going from $\mathbf{a}$ to $\mathbf{b}$ along the path selected is then given by*

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}).$$

This is the fundamental theorem for gradients; it says that the integral (here a *line* integral) of a derivative (here the *gradient*) is given by the value of the function at the boundaries ($\mathbf{a}$ and $\mathbf{b}$).

Geometrical Interpretation: The height of a skyscraper can be measured in two different ways:

1. Measure the height of each floor and add them all up, or
2. Place an altimeter at the top and the bottom, subtract the readings at the ends.

In solved example 1, line integral depended on the *path* taken from **a** to **b**. But the *right* side of

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}).$$

makes no reference to the path - only to the end points. Evidently, *gradients* have the special property that their line integrals are path independent:

Corollary 1: $\int_a^b (\nabla T) \cdot d\mathbf{l}$ is independent of the path taken from a to b.

Corollary 2: $\oint (\nabla T) \cdot d\mathbf{l} = 0$, since the beginning and end points are identical, and hence $T(\mathbf{b}) - T(\mathbf{a}) = 0$.

A conservative force may be associated with a scalar potential energy function, whereas a non-conservative force cannot be expressed in a similar way.

Solved example 4

Let $T = xy^2$, and take point **a** to be the origin (0, 0, 0) and **b** the point (2, 1, 0).

Check the fundamental theorem for gradients.

Let's go out along the x axis (step i) and then up (step ii).

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}; \nabla T = y^2 \hat{\mathbf{x}} + 2xy \hat{\mathbf{y}}.$$

$$(i) \ y = 0; \ d\mathbf{l} = dx \hat{\mathbf{x}}, \ \nabla T \cdot d\mathbf{l} = y^2 dx = 0, \text{ so } \int_i \nabla T \cdot d\mathbf{l} = 0.$$

$$(ii) \ x = 2; \ d\mathbf{l} = dy \hat{\mathbf{y}}, \ \nabla T \cdot d\mathbf{l} = 2xy dy = 4y dy, \text{ so } \int_{ii} \nabla T \cdot d\mathbf{l} = \int_0^1 4y dy = 2y^2 \Big|_0^1 = 2.$$

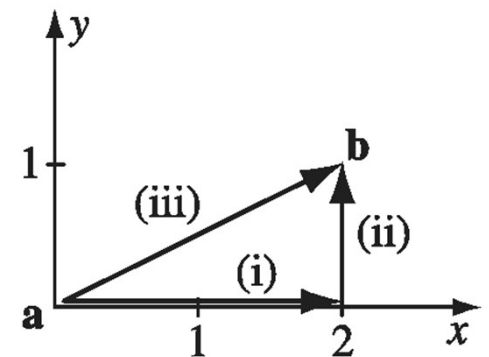
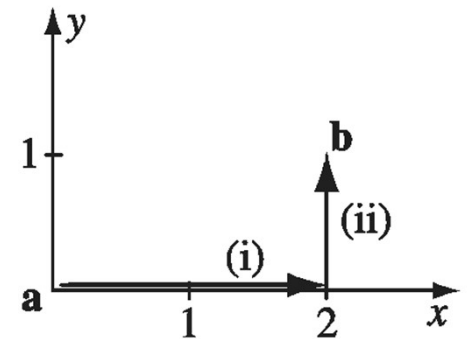
$$\int_i \nabla T \cdot d\mathbf{l} + \int_{ii} \nabla T \cdot d\mathbf{l} = 2 \text{ and } T(\mathbf{b}) - T(\mathbf{a}) = 2 - 0 = 2.$$

What if the path of integration is (iii)?

$$(iii) \ y = \frac{1}{2}x, \ dy = \frac{1}{2}dx, \ \nabla T \cdot d\mathbf{l} = y^2 dx + 2xy dy = \frac{3}{4}x^2 dx, \text{ so}$$

$$\int_{iii} \nabla T \cdot d\mathbf{l} = \int_0^2 \frac{3}{4}x^2 dx = \frac{1}{4}x^3 \Big|_0^2 = 2.$$

The integral is independent of path.



The fundamental theorem for divergences

The fundamental theorem for divergences states that: The integration of a derivative (the divergence) over a region (a volume) is equal to the value of the function at the boundary (the surface that bounds the volume).

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}.$$

This theorem is also known by the names: **Gauss's theorem** and **Green's theorem**.

Geometrical Interpretation: Measures the total amount of fluid passing out through the surface, per unit time. (In general anything that is enclosed within a volume must come out of the surface that encloses the volume).

Solved example 5

Check the divergence theorem using the function $\mathbf{v} = y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + (2yz) \hat{\mathbf{z}}$ and the unit cube situated at the origin.

For the LHS of the divergence theorem,

$$\nabla \cdot \mathbf{v} = 2(x + y),$$

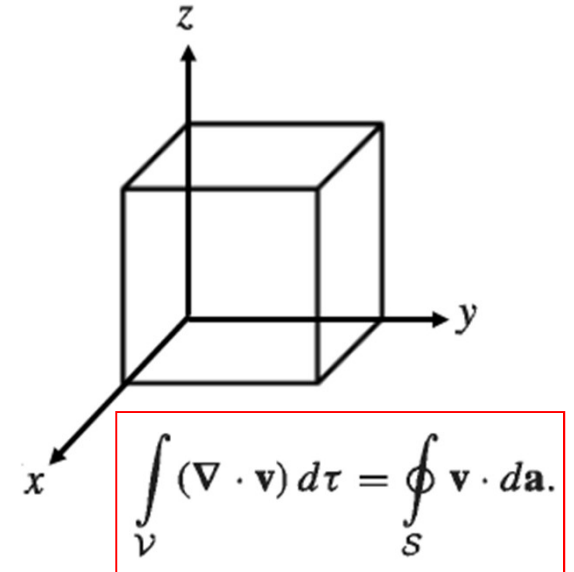
and

$$\int_V 2(x + y) d\tau = 2 \int_0^1 \int_0^1 \int_0^1 (x + y) dx dy dz,$$

$$\int_0^1 (x + y) dx = \frac{1}{2} + y, \quad \int_0^1 (\frac{1}{2} + y) dy = 1, \quad \int_0^1 1 dz = 1.$$

Thus,

$$\int_V \nabla \cdot \mathbf{v} d\tau = 2.$$



(contd...)

For the RHS of the divergence theorem,

$$(i) \quad \int \mathbf{v} \cdot d\mathbf{a} = \int_0^1 \int_0^1 y^2 dy dz = \frac{1}{3}.$$

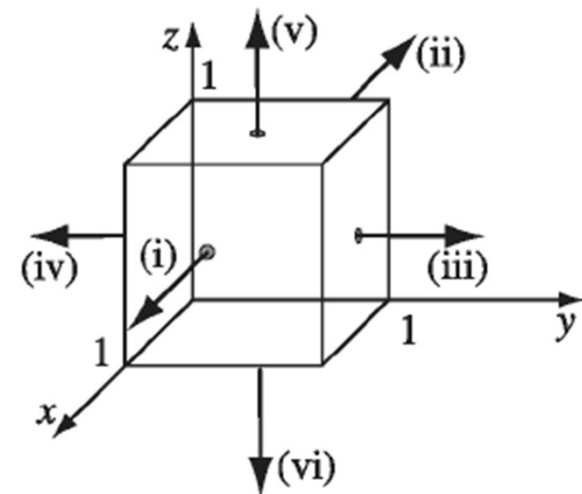
$$(ii) \quad \int \mathbf{v} \cdot d\mathbf{a} = - \int_0^1 \int_0^1 y^2 dy dz = -\frac{1}{3}.$$

$$(iii) \quad \int \mathbf{v} \cdot d\mathbf{a} = \int_0^1 \int_0^1 (2x + z^2) dx dz = \frac{4}{3}.$$

$$(iv) \quad \int \mathbf{v} \cdot d\mathbf{a} = - \int_0^1 \int_0^1 z^2 dx dz = -\frac{1}{3}.$$

$$(v) \quad \int \mathbf{v} \cdot d\mathbf{a} = \int_0^1 \int_0^1 2y dx dy = 1.$$

$$(vi) \quad \int \mathbf{v} \cdot d\mathbf{a} = - \int_0^1 \int_0^1 0 dx dy = 0.$$



The total flux is $\oint_S \mathbf{v} \cdot d\mathbf{a} = \frac{1}{3} - \frac{1}{3} + \frac{4}{3} - \frac{1}{3} + 1 + 0 = 2$

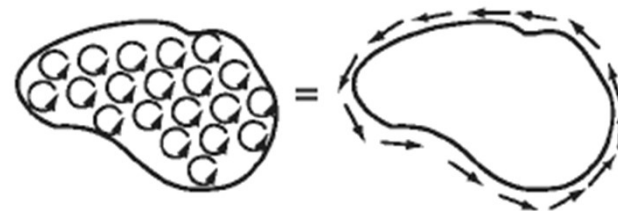
The fundamental theorem for curls

The fundamental theorem for curls (**Stokes' theorem**) states that: The integration of a derivative (here, the curl) over a region (**a patch of surface S**) is equal to the value of the function at the boundary (**the perimeter of the patch P**).

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_P \mathbf{v} \cdot d\mathbf{l}.$$

$\oint \mathbf{v} \cdot d\mathbf{l}$ is sometimes called the circulation of $\mathbf{v}$.

Geometrical Interpretation: The curl measures the "twist" of the vectors $\mathbf{v}$. Now, the integral of the curl over some surface (or, more precisely, the *flux* of the curl *through* that surface) represents the "**total amount of swirl**," and we can determine that just as well by going around the edge and finding how much the flow is following the boundary.



A flux integral depends critically on what surface you integrate over, but evidently this is not the case with curls. For Stokes' theorem says that $(\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ is equal to the line integral of $\mathbf{v}$ around the boundary, and the latter makes no reference to the specific surface you choose.

Corollary 1: $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary lines, not on the particular surface used.

Corollary 2: $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for any closed surface, since the boundary line shrinks down to a point.

These corollaries are analogous to those for the gradient theorem.

Solved example 6

Suppose $\mathbf{v} = (2xz + 3y^2)\hat{\mathbf{y}} + (4yz^2)\hat{\mathbf{z}}$. Check Stokes' theorem for the square surface shown in the figure.

For the LHS of the Stokes' theorem

$$\nabla \times \mathbf{v} = (4z^2 - 2x)\hat{\mathbf{x}} + 2z\hat{\mathbf{z}} \text{ and } d\mathbf{a} = dy dz \hat{\mathbf{x}}. \text{ (Right hand thumb rule.)}$$

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 \int_0^1 4z^2 dy dz = \frac{4}{3}. \quad (\text{here, } x = 0)$$

For the RHS of the Stokes' theorem,

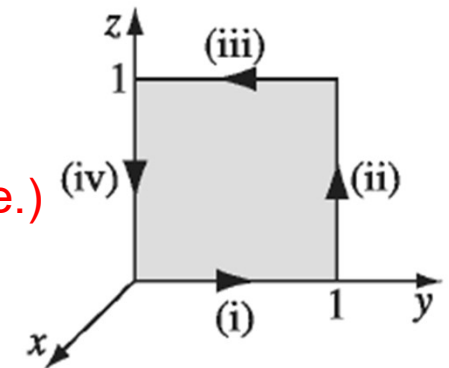
$$(i) \quad x = 0, \quad z = 0, \quad \mathbf{v} \cdot d\mathbf{l} = 3y^2 dy, \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 3y^2 dy = 1,$$

$$(ii) \quad x = 0, \quad y = 1, \quad \mathbf{v} \cdot d\mathbf{l} = 4z^2 dz, \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 4z^2 dz = \frac{4}{3},$$

$$(iii) \quad x = 0, \quad z = 1, \quad \mathbf{v} \cdot d\mathbf{l} = 3y^2 dy, \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_1^0 3y^2 dy = -1,$$

$$(iv) \quad x = 0, \quad y = 0, \quad \mathbf{v} \cdot d\mathbf{l} = 0, \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_1^0 0 dz = 0.$$

$$\oint \mathbf{v} \cdot d\mathbf{l} = 1 + \frac{4}{3} - 1 + 0 = \frac{4}{3}.$$



$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_P \mathbf{v} \cdot d\mathbf{l}$$

Questions?